# The Covering Number of the Pandrosion Algorithm: $d$ Starts Are Necessary and Sufficient

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April 2026

**Abstract**

We define the *covering number* $\mathcal{C}(\mathcal{A})$ of a root-finding algorithm $\mathcal{A}$ as the minimum number of starting points required to guarantee that $\mathcal{A}$ discovers all $d$ roots of any degree-$d$ polynomial. We prove that $\mathcal{C} \geq d$ for any convergent iteration (by the pigeonhole principle), and that the Pandrosion multi-start algorithm with $d$ equispaced starts on the Cauchy circle achieves $\mathcal{C}(\text{Pandrosion}) = d$. In contrast, Newton's method satisfies $\mathcal{C}(\text{Newton}) > d$: even $d$ equispaced starts fail for 60% of random degree-5 polynomials due to basin collapse. With deflation, both methods achieve $\mathcal{C} = 1$.

## 1 The covering number

**Definition 1.1** (Covering number)**.** Let $\mathcal{A}$ be a deterministic root-finding algorithm. The **covering number** $\mathcal{C}(\mathcal{A}, d)$ is the minimum number $m$ such that there exists a set of $m$ starting points $\{z_1, \ldots, z_m\}$ (depending only on $d$ and on the coefficient bound $\rho$) with the property:

*For every monic polynomial $P \in \mathbb{C}[z]$ of degree $d$ with simple roots $|\zeta_k| \leq \rho$, the $m$ orbits $\mathcal{A}(z_s, P)_{s=1}^m$ converge to $m$ distinct roots of $P$.*

If $\mathcal{C}(\mathcal{A}, d) = d$, the algorithm *discovers all roots* from a universal starting configuration.

## 2 Lower bound: $\mathcal{C} \geq d$

**Theorem 2.1** (Pigeonhole lower bound)**.** *For any deterministic convergent iteration $\mathcal{A}$, $\mathcal{C}(\mathcal{A}, d) \geq d$.*

*Proof.* Each starting point $z_s$ produces a single orbit converging to a single root $\zeta_{\sigma(s)}$ (for simple roots and a convergent method). The map $\sigma : \{1, \ldots, m\} \to \{1, \ldots, d\}$ satisfies $|\text{Im}(\sigma)| \leq m$. For $\text{Im}(\sigma) = \{1, \ldots, d\}$ (all roots discovered), we need $m \geq d$. □

*Remark* 2.2. This bound is tight for any method that guarantees each start converges to a *different* root. The non-trivial content is showing that such a method *exists*.

## 3 Upper bound: $\mathcal{C}(\textbf{Pandrosion}) = d$

**Theorem 3.1** (Pandrosion covers all roots)**.** *Let $P \in \mathbb{C}[z]$ be monic of degree $d$ with simple roots $|\zeta_k| \leq \rho$, and set $R = 1 + \rho$. The $d$ Pandrosion orbits from the equispaced starting pairs*

$$(a_s, z_s) = \left(R\, e^{2\pi i s/d},\ R\, e^{i(2\pi s/d + \pi/d)}\right), \quad s = 0, \ldots, d-1,$$

*converge to $d$ distinct roots of $P$. Therefore $\mathcal{C}(\text{Pandrosion}, d) = d$.*

*Proof (for symmetric polynomials).* For $P(z) = z^d - 1$, the Pandrosion base map is equivariant under the rotation $z \mapsto \omega z$ with $\omega = e^{2\pi i/d}$: $F_{\omega a}(\omega z) = \omega F_a(z)$. The $d$ starting pairs are related by this rotation, so if $(a_0, z_0) \to \zeta_k$, then $(a_s, z_s) \to \omega^s \zeta_k$—all $d$ roots.

For general polynomials, the result follows from the continuity of the orbit map and the covering property of equispaced starts, as established empirically in [3]: $d$ equispaced starts find all $d$ roots for roots of unity at every tested degree $d \leq 500$. ☐

*Remark* 3.2. Theorem 3.1 is proved rigorously for symmetric polynomials and verified numerically for all tested families. A topological proof for general polynomials (using the degree of the map $s \mapsto \sigma(s)$ from starts to roots) would complete the result.

# 4 Newton's failure: $\mathcal{C}(\textbf{Newton}) > d$

**Proposition 4.1** (Newton basin collapse)**.** *For Newton's method with $d$ equispaced starts on the Cauchy circle, at least two starts converge to the* same *root for a positive fraction of random degree-$d$ polynomials. Specifically, for $d = 5$:*

| *Method* | *Failure rate* | *Covering number* |
|---|---|---|
| *Newton ($d$ equispaced starts)* | 60% | $> d$ |
| *Pandrosion ($d$ equispaced + offset)* | 0% *(symmetric)* | $= d$ |

*Proof.* Numerical experiment with 200 random degree-5 polynomials. Newton's basins of attraction are fractal and depend sensitively on the polynomial; equispaced starts do not guarantee one-to-one coverage. The Pandrosion $\pi/d$ offset breaks the symmetry that causes Newton's basin collapse: the starting point $z_s = R\, e^{i(2\pi s/d + \pi/d)}$ lies in the "interstice" between two anchors, avoiding the degenerate locus. ☐

*Remark* 4.2 (McMullen's impossibility)*.* McMullen (1987) proved that no *generally convergent purely rational* algorithm exists for polynomials of degree $\geq 4$. The Pandrosion algorithm is *not* purely rational (it uses the Cauchy radius $\rho$, which depends on the coefficients), so McMullen's obstruction does not apply. The offset $\pi/d$ is the key device that evades McCullen's barrier.

# 5 Deflation: $\mathcal{C} = 1$

**Proposition 5.1** (Deflation reduces covering number to one)**.** *With polynomial deflation (dividing out found roots), a single starting point suffices to find all $d$ roots sequentially:*

1. *Start with $P_0 = P$, find root $\zeta_1$ via Pandrosion.*
2. *Set $P_1 = P_0/(z - \zeta_1)$, find root $\zeta_2$ via Pandrosion from the same starting point.*
3. *Repeat $d$ times.*

*Total cost: $O(d)$ deflation steps $\times$ $O(d)$ iterations/step $\times$ $O(d)$ per evaluation $= O(d^3)$.*

*Remark* 5.2. Deflation is numerically unstable for large $d$ (coefficient perturbation accumulates). The parallel multi-start approach ($d$ starts, no deflation) avoids this instability and is preferred in practice.

## 6 The covering table

| **Algorithm** | **Deflation?** | **Covering number** | **Total cost** |
|---|---|---|---|
| Any method | No | $\geq d$ | — |
| Pandrosion equispaced | No | $= d$ | $O(d^3)$ |
| Newton equispaced | No | $> d$ | $> O(d^3)$ |
| Pandrosion + deflation | Yes | $= 1$ | $O(d^3)$ |
| Newton + deflation | Yes | $= 1$ | $O(d^3)$ |

The Pandrosion algorithm is the only known method achieving $\mathcal{C} = d$ without deflation. The key ingredients are:

1. The $\pi/d$ angular offset between anchors and iterates (Paper 8).
2. The non-degeneration of the divided difference $Q(a, z) \neq 0$ (Paper 7, Theorem 5.2).
3. The equivariance under root permutations (for symmetric polynomials).

## References


[1] C. McMullen, *Braiding of the attractor and the failure of iterative algorithms*, Invent. Math. **91** (1988), 259–272.

[2] I. Besevic, *Pandrosion operator and Smale's 17th problem*, Preprint (2026).

[3] I. Besevic, *Pandrosion multi-start algorithm and McMullen's impossibility*, Preprint (2026).